

Kilian Koch



Doctoral Researcher in Mathematics

Chair of Applied Analysis, RWTH Aachen University

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I am a mathematician and doctoral researcher at the Chair of Applied Analysis at RWTH Aachen University. My research concerns gradient flows in geometric analysis, in particular well-posedness and singular behaviour of (half-)harmonic map heat flows, for instance due to topological constraints. My position is funded by the DFG through CRC 1481, project C01.

POSITIONS

April 2024 – Present

Doctoral Researcher

RWTH Aachen University

Chair of Applied Analysis. Funded by the German Research Foundation (DFG) via CRC 1481 *Sparsity and Singular Structures*, project C01.

October 2020 – March 2024

Mathematics Tutor

RWTH Aachen University

Held various tutoring positions, including Preliminary Course Tutor, Analysis 1-3 Tutor, Function Theory I Tutor, and Functional Analysis Tutor.

February 2018 – Present

Software Developer & Customer Service

Auto-Koch, Eschweiler

Developed the customer and vehicle management system KoKi and a scheduling program (desktop applications and locally used websites, JavaScript/Node.js). Regular customer service at the information desk and at exhibitions.

EDUCATION

April 2024 – Present (ongoing)

PhD in Mathematics

RWTH Aachen University

October 2022 – April 2024

M.Sc. Mathematics, with distinction (1.2)

RWTH Aachen University

October 2019 – October 2022

B.Sc. Mathematics, with distinction (1.3)

RWTH Aachen University

August 2016 – August 2019

High School Diploma (Abitur)

Mies van der Rohe School, Aachen

PUBLICATIONS & PREPRINTS

2025

Well-posedness of half-harmonic map heat flows for rough initial data

Kilian Koch, Christof Melcher — arXiv (preprint)

DOI: [10.48550/ARXIV.2504.06933](https://doi.org/10.48550/ARXIV.2504.06933)

TALKS

1 June 2026

Thresholding Scheme for the Half-Harmonic Map Heat Flow
Young Researchers Workshop on Geometric Analysis, Singular PDEs and Numerics, RWTH Aachen University

ORGANIZED EVENTS

1 June 2026 – 3 June 2026

Young Researchers Workshop on Geometric Analysis, Singular PDEs and Numerics (Co-organizer)
Joint workshop of CRC 1481, EDDy and IntComSin (FAU Erlangen-Nürnberg / University of Regensburg), RWTH Aachen University

TEACHING EXPERIENCE

October 2025 – March 2026

Höhere Mathematik 1 für Physiker und E-Techniker | Assistant

RWTH Aachen, Aachen

Professor: Prof. Dr. Ivan Yaroslavtsev

Lecturer: Priv.-Doz. Dr. Olaf Wittich

Colleagues: Valentin Linse, M.Sc., Glal Bacho, M.Sc.

Tasks: Created exercises, held exercise sessions, corrected exams, and prepared exam questions.

August 2025 – September 2025

Mathematics III | Co-lecturer

International Academy RWTH Aachen, Aachen

Lecturer: Prof. Dr. rer. nat. Christof Melcher

Colleagues: Valentin Linse, M.Sc., Glal Bacho, M.Sc.

Tasks: Conducted interactive online exercise sessions for international students, provided academic support and created learning materials.

Partners: RWTH Aachen University, proRWTH-Freunde and Förderer der RWTH Aachen e.V.

April 2025 – September 2025

Partielle Differentialgleichungen 1 | Assistant

RWTH Aachen, Aachen

Professor: Prof. Dr. rer. nat. Christof Melcher

Colleague: Glal Bacho, M.Sc.

Tasks: Created exercises, held exercise sessions, corrected exams, and prepared exam questions.

October 2024 – March 2025

Analysis 3 für Mathematiker | Assistant

RWTH Aachen, Aachen

Professor: Prof. Dr. rer. nat. Christof Melcher

Colleague: Glal Bacho, M.Sc.

Tasks: Created exercises, held exercise sessions, corrected exams, and prepared exam questions.

August 2024 – September 2024

Mathematics III | Co-lecturer

International Academy RWTH Aachen, Aachen

Lecturer: Prof. Dr. rer. nat. Christof Melcher

Colleagues: Georg Heinen, M.Sc., Glal Bacho, M.Sc.

Tasks: Conducted interactive online exercise sessions for international students, provided academic support and created learning materials.

Partners: RWTH Aachen University, proRWTH-Freunde and Förderer der RWTH Aachen e.V.

April 2024 – September 2024

Höhere Mathematik 2 für Physiker und E-Techniker | Assistant

RWTH Aachen, Aachen

Professor: Prof. Dr. Sebastian Noelle

Lecturer: Priv.-Doz. Dr. Olaf Wittich

Colleague: Tom Goertzen, Postdoctoral Research Associate

Tasks: Developed digital exercises, conducted exercise sessions, and graded exams.

RESEARCH FUNDING

- Doctoral researcher position funded by the German Research Foundation (DFG) through CRC 1481 *Sparsity and Singular Structures* (project no. 442047500), project C01 *Singularity formation in dissipative harmonic flows*, since April 2024

AWARDS

- RWTH Dean's List, Winter Semester 2021/22 - Summer Semester 2022
- RWTH Dean's List, Winter Semester 2020/21 - Summer Semester 2021
- RWTH Dean's List, Winter Semester 2019/20 - Summer Semester 2020

SKILLS

- **Programming:** JavaScript/Node.js, C/C++, Python, HTML/CSS; basics of Haskell, Prolog and Java
- **Languages:** German (native), English (C1), Japanese (JLPT N3, $\approx$ B1)